



Samuel Kaz

Design Portfolio

Mechanical Engineer
samuel.kaz1@gmail.com



Background

Education:

- **Stevens Institute of Technology**
 - Master of Engineering: Mechanical Engineering
 - Bachelor of Engineering: Mechanical Engineering
 - Minor in Electrical Engineering

Experience:

- **Mainstream Fluid & Air**
 - Design Engineer
- **H2 Energy Now**
 - (CAD Engineering Intern)
- **PPL (Pennsylvania Power and Light)**
 - (Fiber Engineering Intern)
- **Stevens Engineering and Science Laboratory**
 - Systems and Integration Research and Lab Assistant



Earthworm Inspired Softbody Robot (CAPSTONE Project)

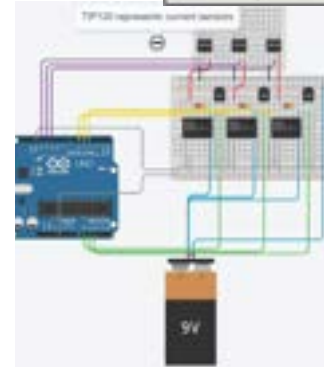
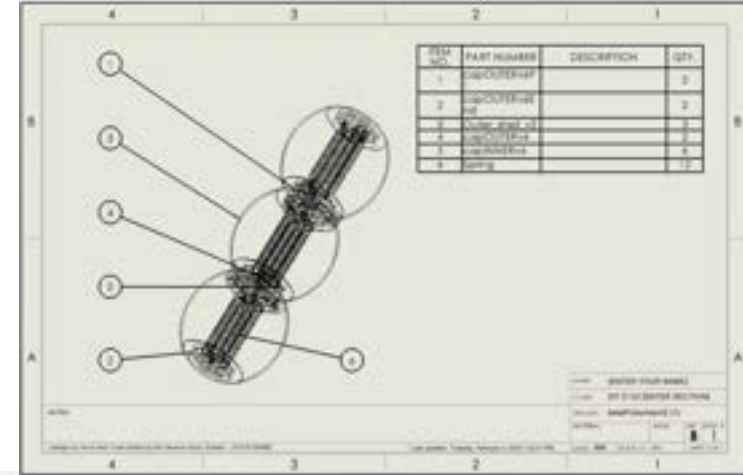
The objective of this project was to create a scaled up earthworm-like soft body robot that could replace the necessity of thrombolysis by creating a vein injectable robot that could be navigated to the blood clot location.

Even though a large prototype was built, it also needed to work within the small blood vessels of the brain. This means that actuation methods such as motors would not work. Shape Memory Alloy (Nitinol) was used.

The senior design team conducted rigorous research in actuation and manufacturing methods, material selection, and conducted technical analysis.

Developed multiple production phases and presented scheduled reports and 2D Drawings including BOM documenting the project's successful progression. Segment were able to expand and contract, moving 5 mm per cycle

Skills Implemented: 3D Modeling, Arduino, 3D Printing, Technical Analysis, Prototyping, Teamwork



Automatic Pet Feeder

- Collaborated with a small team to design an automatic food, water and toy dispensing robot.
- Programed microcontrollers to regulate different actuating systems based on changes in weight, pressure, and movement within its environment.
- Utilized MATLAB Simulink for electrical component selection and to design the robot's circuitry.

Skills Implemented: Solidworks, Simulink, Assembly, Prototyping, Electronics, Teamwork

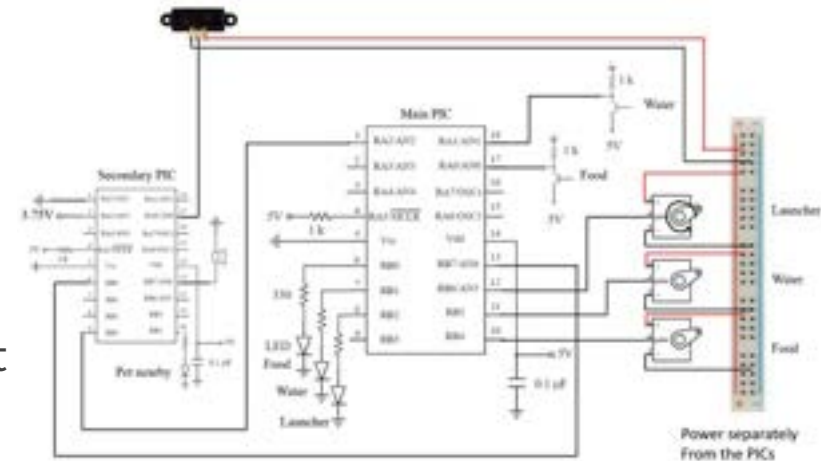


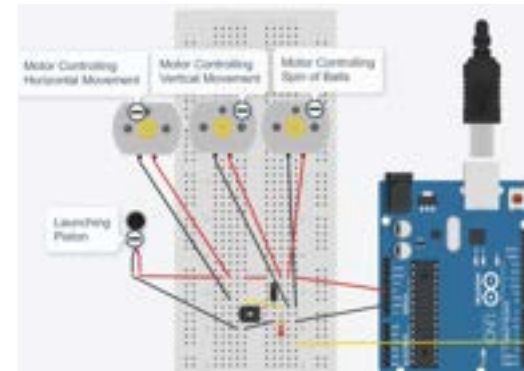
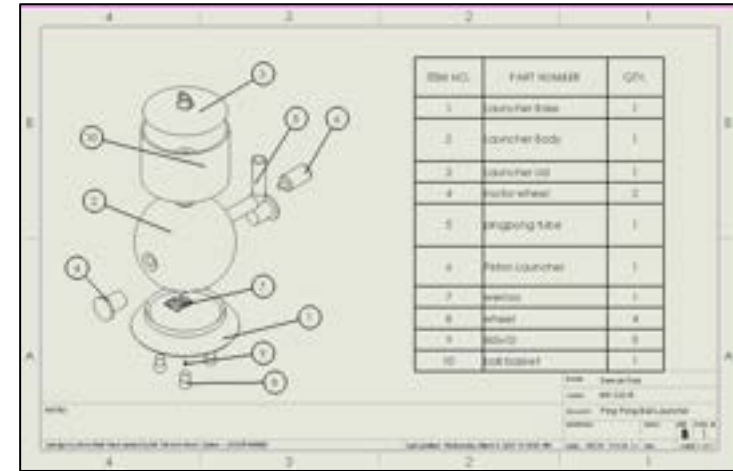
Table Tennis Ball Launcher

Developed a prototype ball launcher that shoots ping pong balls in constant intervals and included the following:

- Interface that controls launching speed, direction, and height
- Storage container that feeds directly into launching mechanism
- Net attachment for ball collection

This project aims to create an alternative to conventional machines that balances thorough & effective features with an affordable price

Skills Implemented: Solidworks, Arduino, 3D Printing, Concept Selection, Electronics, C++, Teamwork

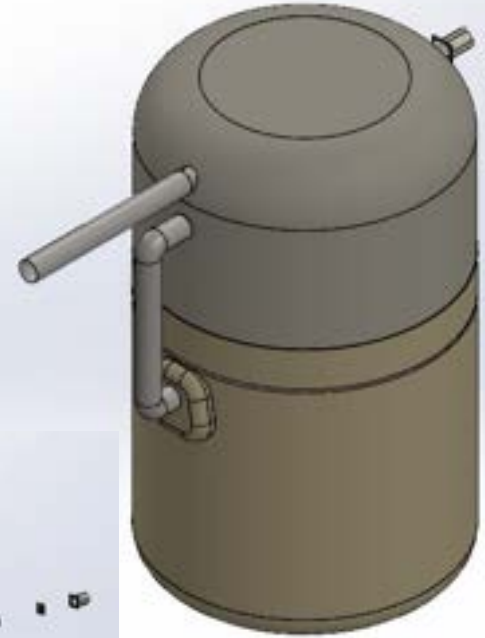
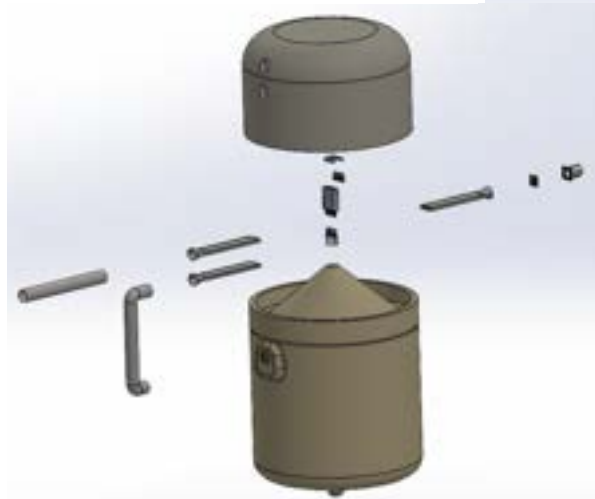


Separation Chamber Housing (Internship)

3D modeled a cooling and filtration enclosure, and storage container for a hydrogen separation chamber designed to output 100 kg of hydrogen per day

Conducted research on gas separation, hydrogen filtration, and heat exchangers to be used in the housing, designing multiple iterations for water flow through the chamber,

Additionally attended meetings and aided in decision making regarding materials and design methods for the patented hydrogen separation chamber.



Autonomous Navigation Robot

- Constructed an autonomous robot system that navigated around an area through obstacles using LIDAR and ultrasonic sensors for pinpointing the location.
- Developed and implemented pathfinding and object avoidance algorithm using C++ and Arduino
- Utilized SolidWorks and 3D printing to model and print the mechanical chassis to support the subsystems

*Skills Implemented: Solidworks, Arduino, 3D Printing,
Electronics, C++, Teamwork*

